SHALINI BANSAL

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SUMMARY

Mechanical engineer with 4+ years of hands-on machining, zero-to-one prototyping, and R&D hardware development. Experienced in programming and operating multi-axis CNC machines, including 5-axis simultaneous milling and live tooling on lathes holding tolerances to ±0.0005”. Skilled at bridging the gap between abstract design and physical hardware while providing DFM guidance.

EDUCATION

Master of Science in Computer Science, *Sofia University, Irvine, California* — *Jan 2025 – Present*
Master of Science in Mechanical Engineering, *University of California, Riverside, California* — *Jan 2021 – Dec 2021*
Bachelor of Engineering in Mechanical Engineering, *Visvesvaraya Technological University, India* — *Aug 2016 – Sep 2020*

WORK EXPERIENCE

Associate Mechanical Engineer, *Haas Automation Inc., Oxnard, CA* — *March 2026 – Present*

- Led end-to-end redesign of chip conveyor mechanisms inherited mid-cycle. Evaluated designs, identified possible failures, and drove assembly to production-ready specs through iterative design and DFM review.
- Redesigned coolant sanitizer assembly from an inherited NPI concept, scoped requirements, performed root cause analysis on leakage failures, and iterated on mechanism design to meet manufacturing and usability targets.
- Contributed to cross-functional design reviews, driving design decisions across the mechanical engineering department.
- Presented chip conveyor design and application guidance at the Haas Dealer Meeting, directly influencing purchasing decisions across dealer networks.

Machine Test Engineer 2, *Haas Automation Inc., Oxnard, CA* — *Jun 2022 – March 2026*

- Set up and operated 5-axis CNC mill, lathe, and mill-turn machines for R&D validation parts, programmed simultaneous 5-axis toolpaths in Fusion 360 CAM, holding tolerances to ±0.0005" on aluminum, copper, and steel.
- Designed and machined custom test components and fixtures in Fusion 360, from CAD through toolpath generation and manual G-code programming, to diagnose machine issues and validate software and hardware features.
- Built and maintained custom workholding solutions for complex setups, established optimized CAM tool libraries, and developed structured first-article inspection workflows to verify machined parts against engineering specs.

Graduate Research Assistant, *University of California, Riverside*, *CA* — *Jan 2021 – Dec 2021*

- Fabricated electromechanical prototypes for a steerable robotic catheter in direct collaboration with a research professor, taking designs from concept to physical hardware using SLA resin 3D printing and applying GD&T and DFM throughout.
- Identified design limitations in inherited concepts, implemented iterative changes to improve steering performance and structural integrity, and conducted motion tests to validate each design revision.

Mechanical Engineering Intern, *Hindustan Aeronautics Limited*, *India* — *Feb – Jun 2020*

- Optimized machining parameters on a 5-axis CNC mill for an aircraft engine blisk and verified surface roughness of the machined prototype against engineering specs.

TECHNICAL SKILLS

CAD/CAE: SolidWorks, CATIA V5, Siemens NX, Creo, Fusion 360, AutoCAD, FEA, Onshape, GD&T (ASME Y14.5)

Machining: 5-Axis CNC Machining, Mill-Turn, Live Tooling, Fusion 360 CAM, G-code, Workholding Design, First-Article Inspection, Sheet Metal, SLA 3D Printing, DFM/DFA

Software: Python, SAP, ECTR, TargetProcess, Jira, Perforce

LEADERSHIP & COMMUNITY

- Designed and ran a training program for all incoming engineers, covering machining workflows and hands-on CNC machining.
- Represented Gene Haas Foundation at the FIRST Robotics World Championship, leading technical demonstrations of CNC manufacturing technology and engaging with student engineering teams.
- Mentored two undergraduates on graduate school applications, research direction, and career planning; led onboarding discussions for 300+ incoming undergraduates at UCR.
- Captained VTU’s Women’s Softball and Handball teams to inter-university championship titles.